GRASYA ENTERPRISE MANAGER

DRAFT USER MANUAL

Version 2.2.0

Document status: Draft for review
Supported desktop platforms: Windows and Linux

G**RASYA** is a database migration and Real-Time Change Data Capture (CDC) platform built specifically for moving data from **PostgreSQL** to **MongoDB**. It's designed to handle not just a one-time data dump, but a full migration lifecycle — from schema analysis to live replication to verification.

Core capabilities:

- **Automated relation extraction** — discovers foreign key relationships directly from the source database
- **Topological dependency sorting** — determines the correct order to migrate tables so dependencies are respected
- **Debezium/Kafka stream processing** — captures database changes as a stream of events
- **CDC mirroring** — keeps MongoDB in sync with PostgreSQL in near real-time after the initial migration
- **Custom calculated fields evaluation** — recomputes generated/derived columns that Postgres wouldn't stream naturally
- **Merkle tree verification** — cryptographically confirms that source and target data actually match

In short: GRASYA doesn't just copy tables once — it maps relational data into a document model, streams ongoing changes, and mathematically verifies the two databases stay consistent.

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1. PURPOSE AND INTENDED USERS

1.1. Purpose of GRASYA

GRASYA Enterprise Manager helps move data from a PostgreSQL source database to a MongoDB target database. It is designed for more than a one-time copy. After copying the existing rows, GRASYA can continue applying new inserts, updates, and deletes from PostgreSQL to MongoDB until the migration is ready to be completed.

GRASYA also helps convert relational data into a MongoDB document structure. Foreign-key relationships discovered in PostgreSQL can be represented as embedded documents or as references between MongoDB collections. Before a migration is completed, GRASYA can verify the source and staged target data and report schema changes that may affect consistency.

The normal migration flow is:

1. Connect to PostgreSQL and MongoDB.
2. Review the tables detected in the selected PostgreSQL schema.
3. Configure relationship mappings and calculated fields.
4. Start the initial snapshot and transformation.
5. Monitor live change data capture (CDC).
6. Verify the data and finish the migration.

1.2. Intended users

This manual is intended for:

- Database operators responsible for a PostgreSQL-to-MongoDB migration.
- Database administrators who can provide connection and replication access.
- Migration administrators who decide how relational data should be represented in MongoDB.
- Technical users responsible for monitoring and completing a migration.

Users should understand basic database terms such as host, port, database, schema, username, and collection. Knowledge of Kafka or Debezium is not required.

1.3. Provider-managed services

GRASYA uses provider-managed cloud services to transfer and process database changes. These services are configured and maintained by the GRASYA provider. The client is responsible for supplying reachable PostgreSQL and MongoDB connections and the required database permissions.

NOTE: The Monitor screen displays Kafka as the managed event bus. Its presence in the diagram does not mean that the user must install or manage Kafka.

2. CLIENT SYSTEM AND NETWORK REQUIREMENTS

2.1. Supported application packages

GRASYA Enterprise Manager is distributed as a desktop application for:

- Windows: installer package or portable application.
- Linux: AppImage package.

Use the application package supplied by your organization or the GRASYA provider. Do not download installers from an unverified source.

2.2. Client computer requirements

The client computer should have:

- A supported 64-bit Windows or Linux operating system.
- Sufficient permission to install or run the provided desktop application.
- A reliable internet connection to the GRASYA cloud service.
- Correct system date and time, especially when TLS certificates are used.
- Access to any organization-required VPN or private network.

2.3. Database network requirements

The GRASYA cloud service must be able to reach both databases. It is not enough for PostgreSQL or MongoDB to be reachable only from the user's desktop.

Before opening the migration wizard, confirm the following with the database or network administrator:

- The PostgreSQL hostname and port are reachable from the GRASYA cloud service.
- The MongoDB hostname and port are reachable from the GRASYA cloud service.

- Required firewall rules, trusted-IP lists, private routes, or VPN connections are active.
- TLS certificates and connection requirements are valid.
- The database accounts are allowed to connect from the GRASYA service.

IMPORTANT: A value such as localhost or 127.0.0.1 normally refers to the computer on which a service is running. Do not use it for a hosted migration unless the GRASYA provider has supplied a tunnel or other approved connection Method.

2.4. Information to collect

PostgreSQL source:

- A descriptive source name.
- Hostname or IP address.
- Port number, commonly 5432.
- Database name.
- Schema name, commonly public.
- Login username and password.
- Whether SSL is required.
- Replication username and password, if different from the login account.
- Provider- or administrator-supplied capture settings.

MongoDB target:

- MongoDB connection URI.
- Target database name.
- Any TLS options required in the URI.
- A MongoDB account with the required access to the target database.

SECURITY NOTE: Store credentials in an approved password manager. Do not place passwords, private connection URIs, or secrets in this manual or in screenshots.

3. INSTALLING AND STARTING GRASYA

3.1. Installing on Windows

Installer package:

1. Obtain the GRASYA Enterprise Manager Setup from an approved source.
2. Open the installer.

3. Follow the installation prompts.
4. Start GRASYA Enterprise Manager from the Start menu or desktop shortcut.

If Windows displays a security prompt, confirm that the publisher and file source matches the information supplied by your organization. If they do not, cancel the installation and contact support.

Portable package:

1. Obtain and extract the portable package if it is distributed as an archive.
2. Place it in a folder where the user has permission to run applications.
3. Open the GRASYA Enterprise Manager executable.

The portable package runs without the normal installer workflow, but the user may still need permission from the organization's endpoint-security system.



Figure 1. Installing GRASYA Enterprise Manager on Windows.

3.2. Starting on Linux

GRASYA is distributed for Linux as an AppImage.

1. Download the approved AppImage package.
2. Mark the file as executable using the desktop file-properties window or the method approved by your organization.
3. Open the AppImage.
4. If the desktop environment requests confirmation, verify the file source before continuing.

[PHOTO HERE — Figure 2: GRASYA AppImage file and Linux launch screen]
Suggested caption: Figure 2. Opening the GRASYA Enterprise Manager AppImage on Linux.

3.3. Main application window

After GRASYA opens, the main window displays:

- A toolbar containing connection and migration controls.
- A navigation area for the available database objects.
- A tabbed workspace beginning with the Start Page.
- A status bar at the bottom of the window.

The Connect control is available when there is no active database connection. Controls such as Run, Pause, Verify, and Finish become available only when their required connection or migration phase has been reached.

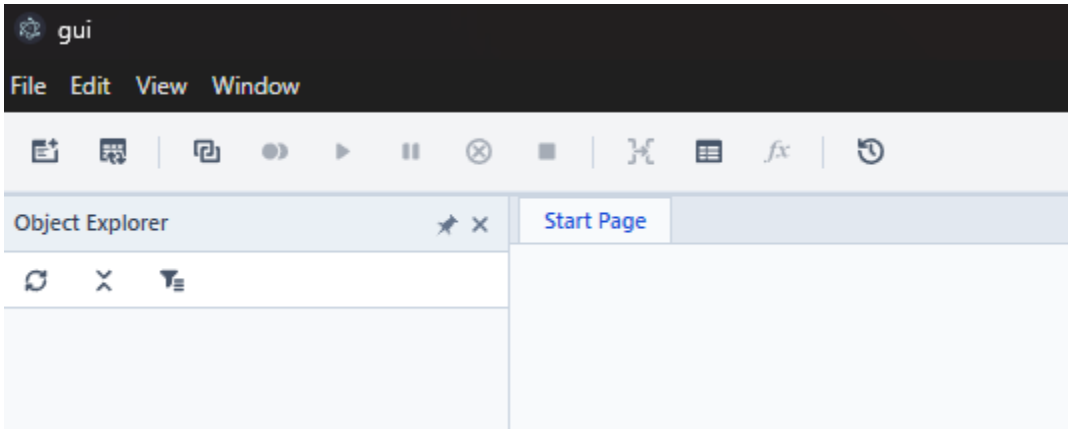


Figure 3. Main application window with its navigation areas before connecting to a migration source and target.

3.4. Cloud-service status

GRASYA creates a cloud session when the application starts. The bottom status bar reports whether that service is available.

- Disconnected means no PostgreSQL/MongoDB migration connection is active.
- Ready means the source and target were successfully registered for the current session.
- No connection — attempting to reconnect means the application temporarily cannot reach the GRASYA cloud service and will retry.
- No connection to server. Try again later means the automatic retry limit was reached. Select Retry after checking the network connection.



Figure 4. Connection information appears in the status bar at the bottom of the application.

4. PREPARING POSTGRESQL AND MONGODB

4.1. PostgreSQL source

Ask the PostgreSQL administrator to confirm that:

- The database server supports logical replication.
- Logical replication is enabled according to the organization's database platform requirements.
- The supplied login can connect to the database and read the selected schema.
- The replication account has the permissions needed to create or use the configured publication and replication slot.
- The tables to migrate have appropriate primary keys. Primary keys are important for stable row identity and verification.
- The selected schema contains the tables intended for migration.
- The source host, port, and SSL settings are correct.

The application validates the PostgreSQL connection when the user advances from the first wizard page. It also detects all tables in the chosen schema.

4.2. MongoDB target preparation

Ask the MongoDB administrator to confirm that:

- The connection URI is valid and reachable from the GRASYA cloud service.
- The URI includes the required authentication and TLS options.

- The MongoDB account can create, read, update, and delete data in the target.
- The intended target database name has been approved.
- The target has adequate storage for staging data and migrated collections.

WARNING: The Cancel command can permanently drop the complete target MongoDB database configured for the migration. Use a dedicated migration database. Do not select a database containing unrelated or valuable data.

4.3. Provider-managed components

No Kafka or Debezium installation is required on the client computer. The user should not create topics, install connectors, or manage those services. If the Capture Settings values are not already known, request them from the database administrator or GRASYA service administrator.

5. CONNECTING THROUGH THE MIGRATION WIZARD

Select Connect on the toolbar to open the four-step Migration Wizard. The Next button validates each database connection before moving forward. Back returns to an earlier page. Cancel closes the wizard without beginning a migration.

5.1. Step 1: Source Connection

Enter the PostgreSQL source details:

- Source Name: a descriptive name used to identify the source in history.
- Host / Server: the hostname or IP reachable by the GRASYA cloud service.
- Port: the PostgreSQL port, commonly 5432.
- Database Name: the PostgreSQL database containing the source tables.
- Schema: the schema to migrate, commonly public.
- Username and Password: credentials for the source connection.
- Use SSL for source connection: enable this when the PostgreSQL server requires an encrypted connection.

Select Next. GRASYA validates the connection and automatically detects every table in the selected schema. If validation fails, remain on this page and use the displayed error to correct the connection information.

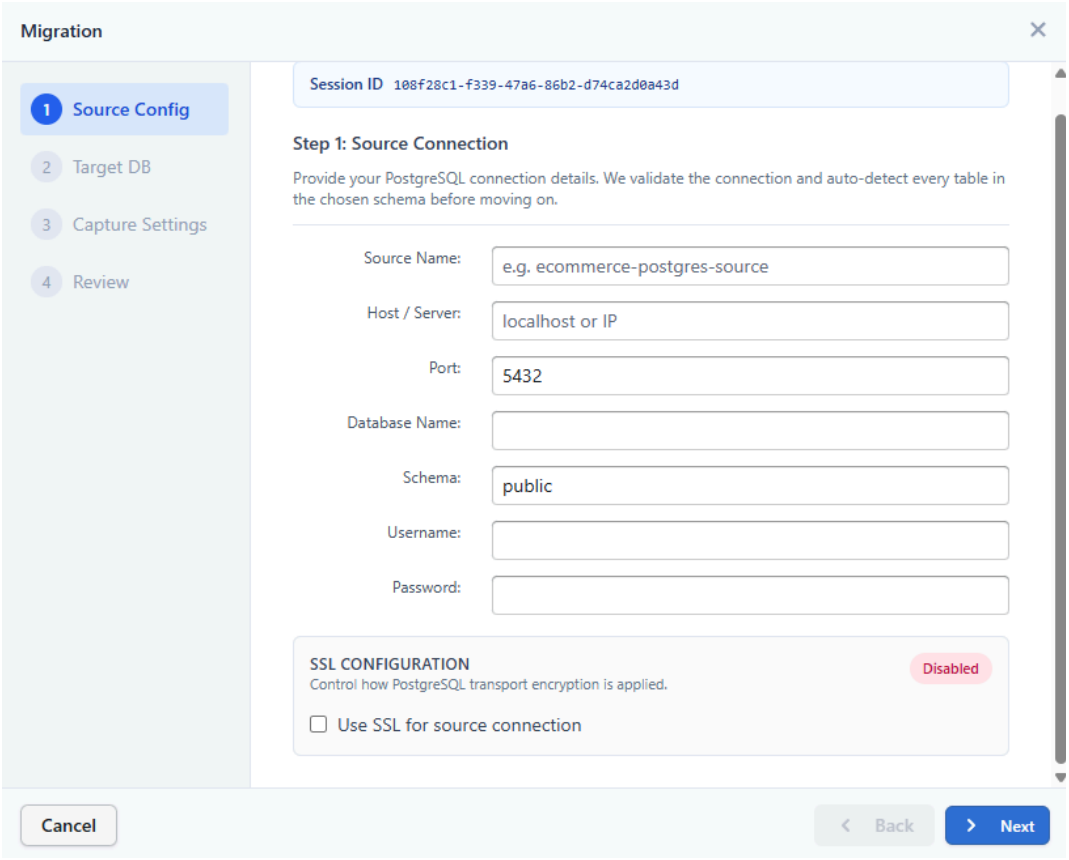


Figure 5. Entering the PostgreSQL source connection.

5.2. Step 2: Target Destination

Enter the MongoDB target details:

- MongoDB URI: the full connection URI, including authentication and TLS options where required.
- Target DB Name: the dedicated MongoDB database to receive the migration.

Examples displayed by the application indicate that a standard mongodb URI or a mongodb+srv URI may be used. Always use the exact URI issued by the MongoDB administrator.

Select Next. GRASYA tests the MongoDB connection before continuing.

WARNING: Carefully check the target database name. Cancel Migration can delete this entire database.

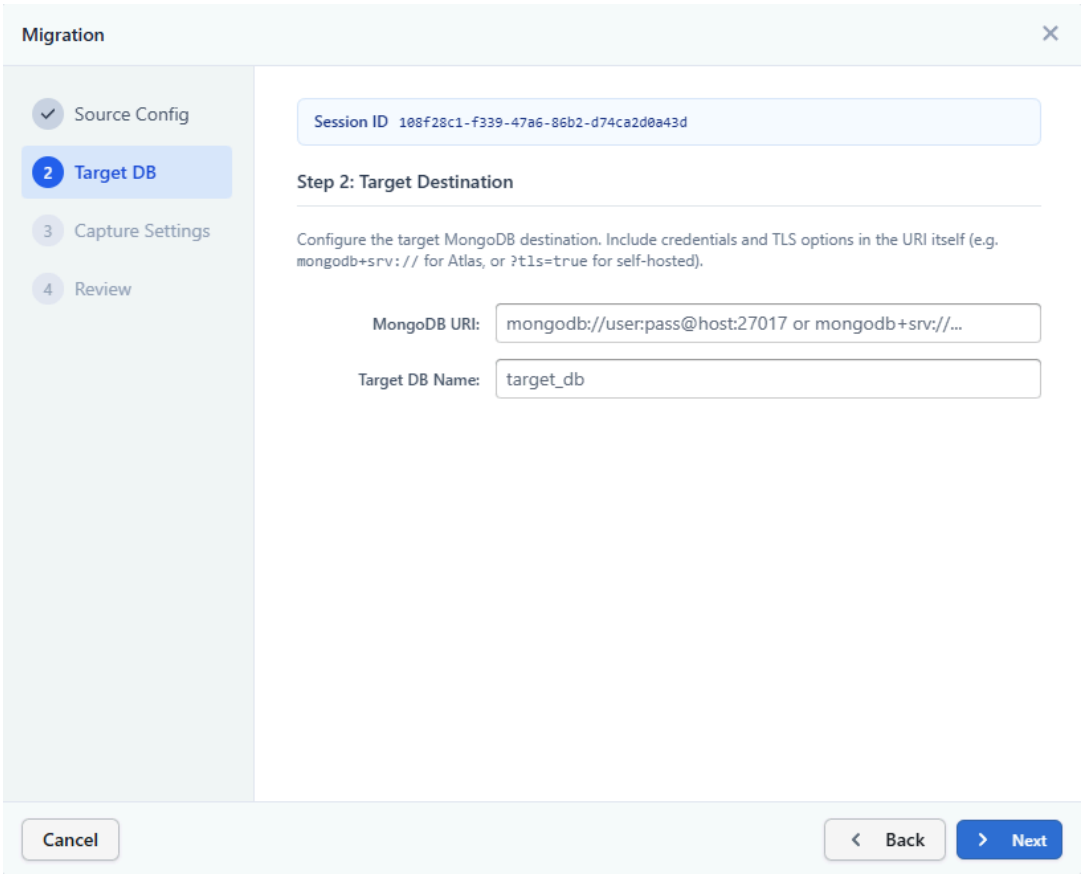


Figure 6. Entering the MongoDB destination.

5.3. Step 3: Capture Settings

This page contains settings used by the provider-managed change-capture service. Keep the supplied or prefilled values unless a database administrator or GRASYA service administrator has instructed you to change them.

- Topic Prefix: identifier used by the managed migration service.
- Included Schema: schema captured from PostgreSQL. This should match the source schema unless an administrator specifies otherwise.
- Replication User and Replication Password: PostgreSQL credentials authorized for logical replication.
- Slot Name: stable name used to track the source change stream. Concurrent migrations against the same PostgreSQL source must use distinct slot names.
- Publication Name: PostgreSQL publication used for captured changes.
- Snapshot Mode:
  - initial copies the existing PostgreSQL rows before entering live CDC. Use this for a normal full migration.

- no_data skips the initial copy and listens only for new changes. Use this only when the target has already been prepared and an administrator has explicitly instructed you to perform a CDC-only run.

IMPORTANT: For most new migrations, select initial. Choosing no_data for an empty target will not copy existing PostgreSQL rows.

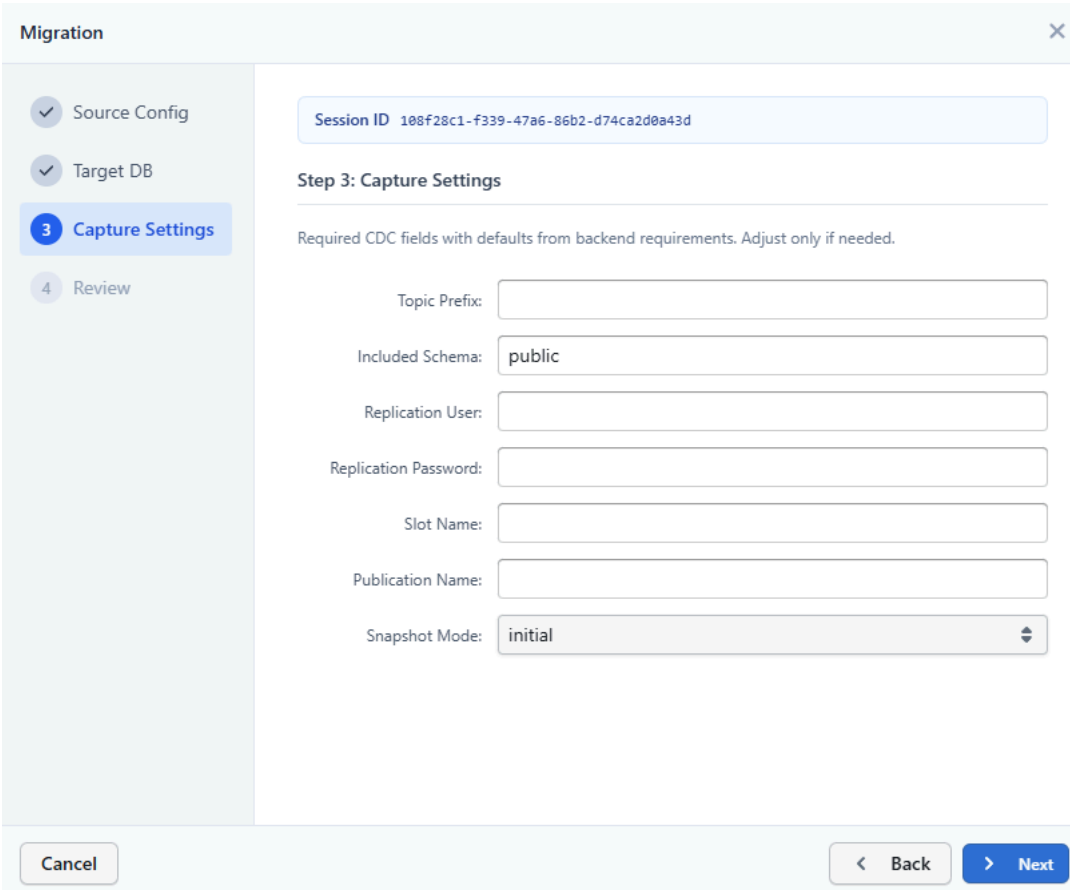


Figure 7. Provider-guided change-capture settings.

5.4. Step 4: Review

The Review page summarizes:

- PostgreSQL endpoint, schema, user, and SSL choice.
- MongoDB URI and target database.
- Capture settings.
- Number and names of detected tables.

Review the values carefully. In particular, verify the MongoDB target database, snapshot mode, PostgreSQL schema, and detected table list. If information is incorrect, select Back and correct it.

Select Finish to register the migration configuration and return to the main workspace. Finish completes the connection wizard; it does not start copying data. Configure relationships and calculated fields before selecting Run.

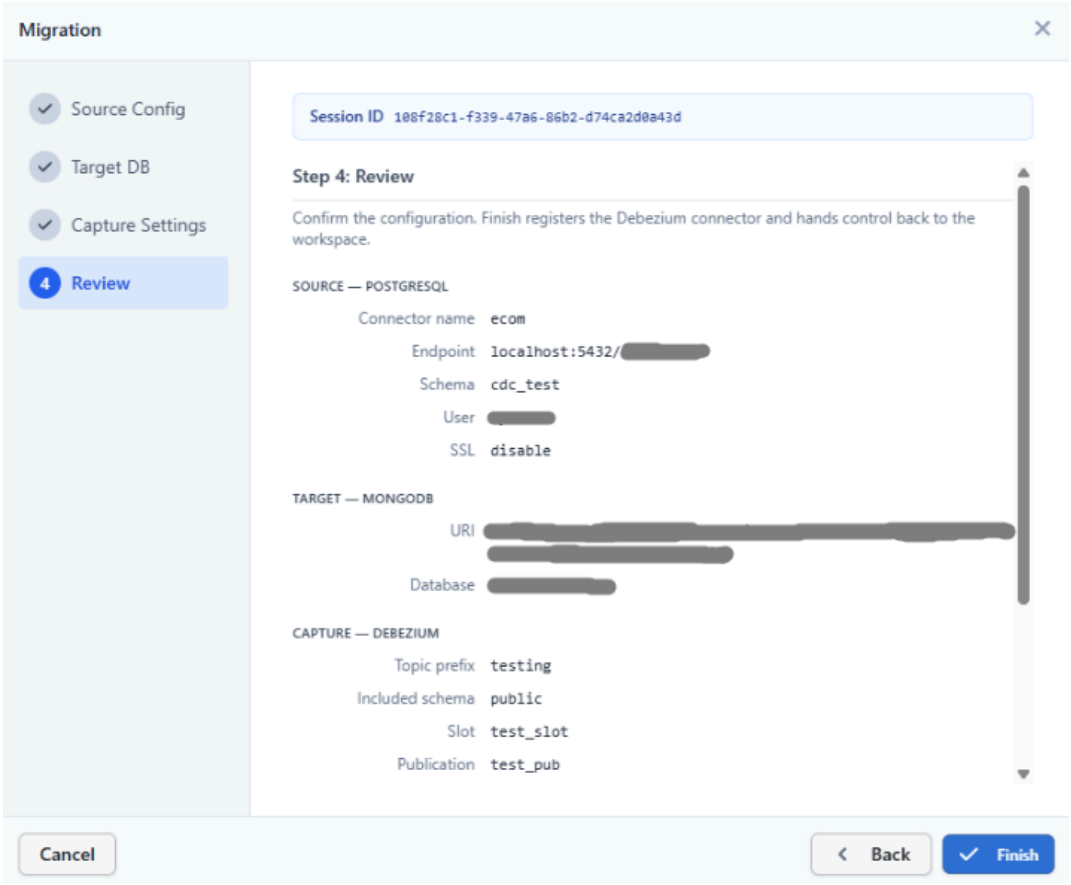


Figure 8. Reviewing the source, target, capture settings, and automatically detected tables before finishing the wizard. Some sensitive details are censored.

6. REVIEWING TABLES AND CONFIGURING RELATIONSHIPS

6.1. Detected tables

The current wizard automatically detects and includes every table in the selected PostgreSQL schema. The Review page lists those tables, but it does not provide individual table-selection checkboxes.

If unwanted tables appear, return to Source Connection and choose the correct schema. If the migration requires only a subset of tables, contact the GRASYA administrator before starting the migration rather than assuming that a table can be excluded from the current interface.

6.2. Entity-relationship diagram

After the wizard finishes, GRASYA loads the schema into the ERD Diagram tab. Tables are represented as objects containing their columns, and connecting lines represent detected PostgreSQL foreign keys.

Use the ERD to confirm that the expected tables and relationships were found. Selecting or hovering over a relationship highlights its corresponding entry in the relationship-mapping panel.

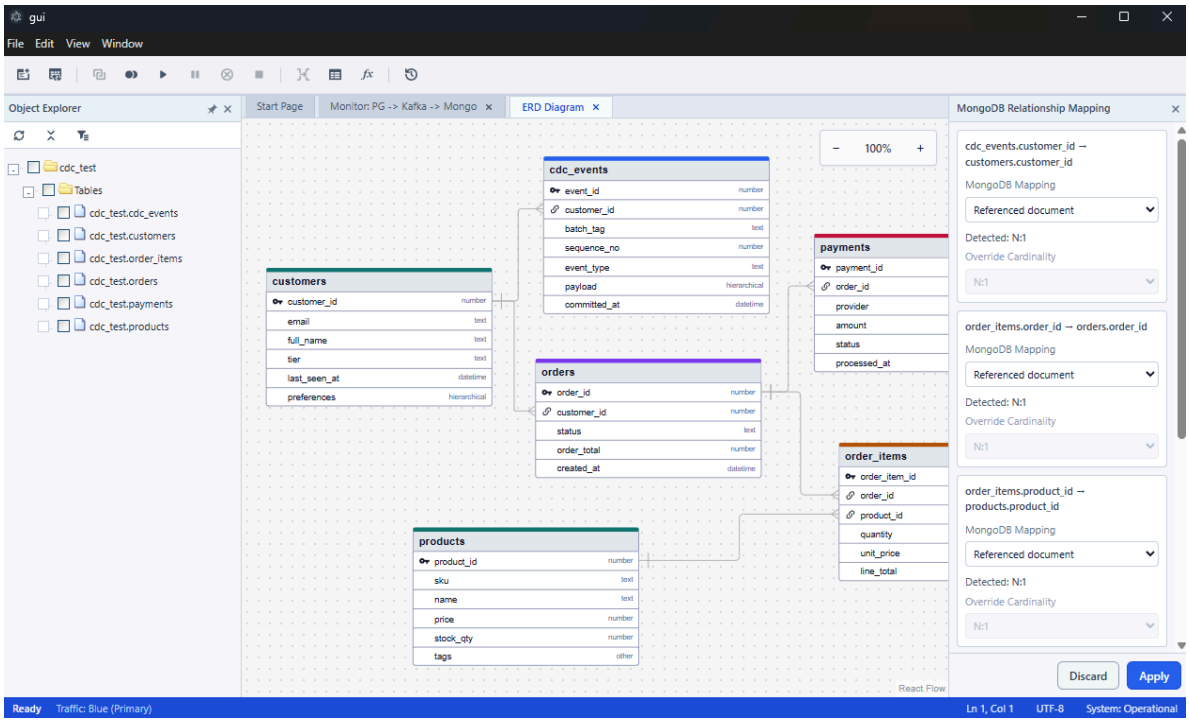


Figure 9. The ERD displays tables, columns, and detected foreign-key relationships from the selected schema.

6.3. Embedded and Referenced mappings

Open MongoDB Relationship Mapping from the toolbar. For each detected relationship, choose one of the following:

- Embedded document: child data is placed inside the related parent document. This is useful when the child data is normally read together with its parent.
- Referenced document: related data remains in a separate MongoDB collection and is linked through a reference. This is the default choice and is useful when records have independent use or may grow significantly.

GRASYA displays a detected cardinality such as 1:1, 1:N, N:1, or N:N. A cardinality override is available for Embedded mappings. The override is disabled for Referenced mappings because the referenced representation is handled separately by the transformer.

Select Apply to save all displayed relationship choices to the current session. Discard does not apply the current pending choices; it does not remove relationship mappings that were applied earlier. Confirm the displayed selections before leaving the panel.

Relationship mappings cannot be applied while a migration is running. Make these decisions after the wizard and before selecting Run.

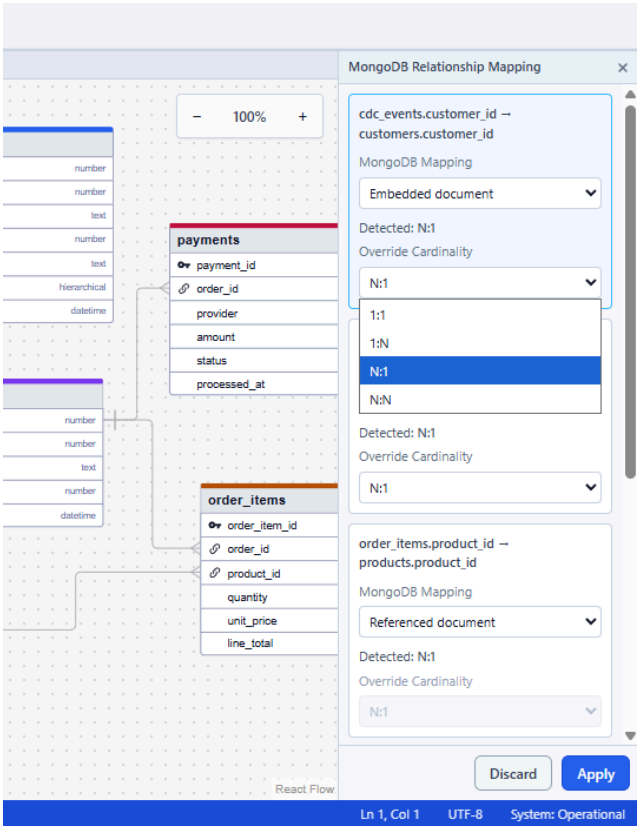


Figure 10. Choosing Embedded or Referenced representation and reviewing the detected relationship cardinality.

7. CONFIGURING CALCULATED FIELDS

Calculated fields allow GRASYA to recompute generated or derived values that may not be included in PostgreSQL live-change events.

7.1. Opening the panel

After connecting and before starting the migration, select Calculated Fields from the toolbar. The panel loads:

- Previously saved calculated fields for the current session.
- Generated PostgreSQL columns detected by GRASYA.
- Suggested expressions for detected generated columns.

Detected suggestions are enabled by default. The backend also registers valid generated-column suggestions at migration start as a safety measure, but the user should still review this panel before running the migration.

7.2. Managing fields

Each enabled row contains:

- Table name.
- Column name.
- Calculation expression.
- An optional display of the original PostgreSQL expression.

Use the checkbox to include or exclude a row. Select + Add field to create a manual calculated field. Use the trash control on a row to remove it from the pending configuration.

Expressions run in a restricted environment. The interface supports basic arithmetic and comparisons and lists these functions:

- lower
- upper
- round
- abs
- len
- min

- max
- concat
- coalesce

Select Apply to validate and save the enabled fields. A rejected expression is highlighted with its error. Correct the table, column, or expression and apply again.

Calculated fields cannot be changed while a migration is running.

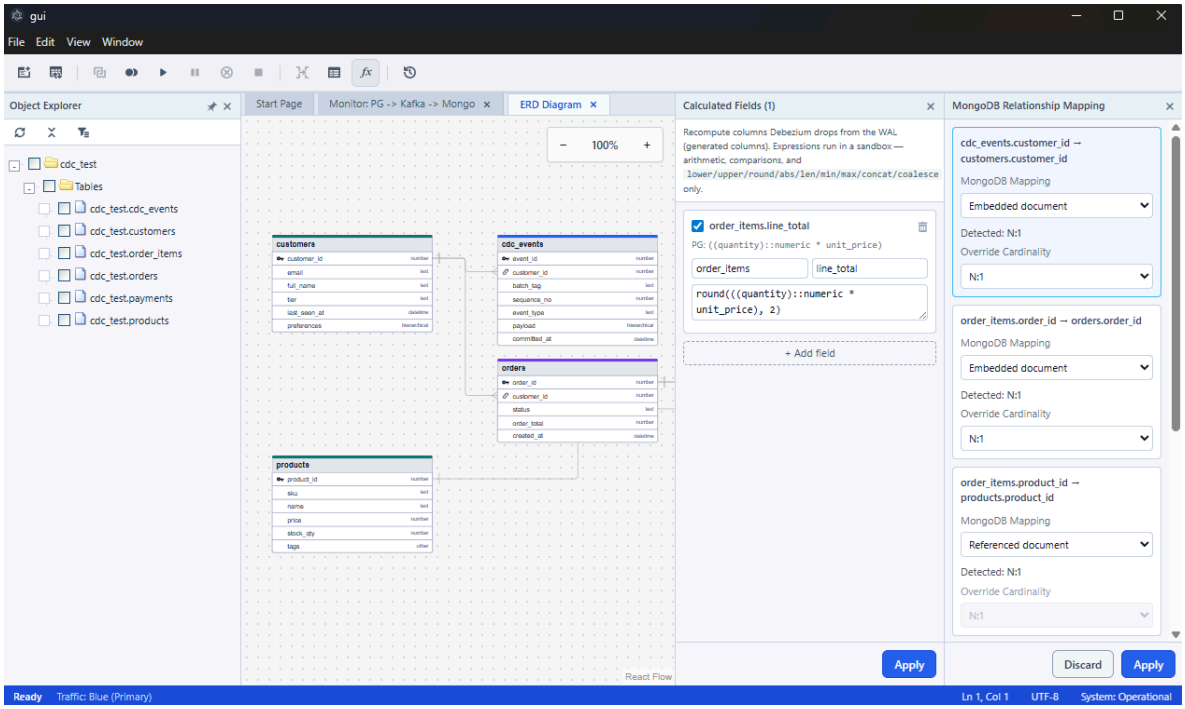


Figure 11. Reviewing an auto-detected generated column and its suggested calculated-field expression.

8. STARTING AND MONITORING A MIGRATION

8.1. Starting the migration

Before selecting Run, confirm that:

- The correct PostgreSQL and MongoDB databases are connected.
- The detected table list is correct.
- Relationship mappings have been applied.
- Calculated fields have been reviewed and applied.
- The target database is dedicated to this migration.

Select Run on the toolbar. GRASYA determines the table dependency order, starts the migration worker, and opens the Monitor tab.

8.2. Migration phases

A normal migration using initial snapshot mode progresses through:

1. Snapshot and staging: existing PostgreSQL rows are copied into a staging area in MongoDB.
2. Verification gate: GRASYA checks for schema changes and compares the source rows with staged data.
3. Transformation: staged rows are converted into migrated collections using the configured relationship mappings and calculated fields.
4. CDC: live PostgreSQL inserts, updates, and deletes continue to flow to MongoDB until the migration is finished.

The logs may identify the phases as Phase1a, Phase1b, and Phase2. Phase2 is the live CDC phase in which Verify and Finish Migration become available.

8.3. Monitor diagram

The Monitor view shows PostgreSQL as the source, Kafka as the provider-managed event bus, and MongoDB as the target. Animated activity indicators show that a migration is running. Users do not need to operate the Kafka component shown in this diagram.

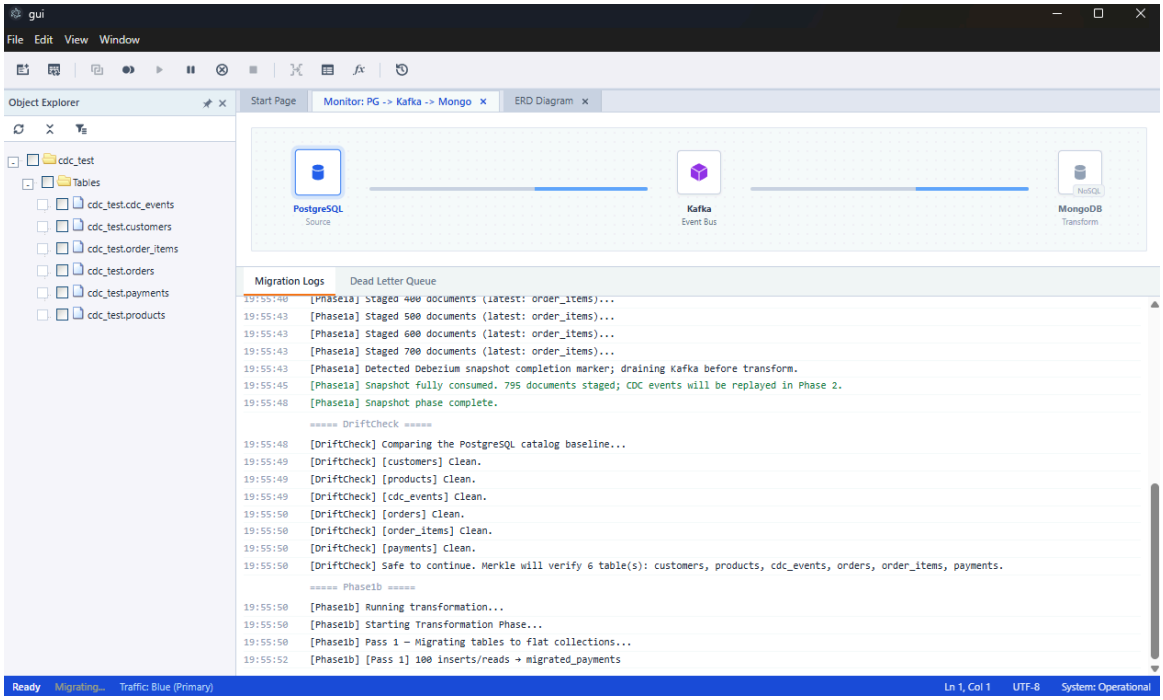


Figure 12. Live migration activity from PostgreSQL through the managed event bus to MongoDB.

8.4. Reading migration logs

The Logs tab displays timestamped messages from the application and backend. Use the message type and color to distinguish:

- Information: a normal operation or phase update.
- Success: an operation completed successfully.
- Warning: attention is needed, but the migration may still continue.
- Error: an operation failed or requires correction.

Keep the Monitor open when starting, verifying, or finishing a migration. The logs provide the most direct explanation when an operation does not complete.

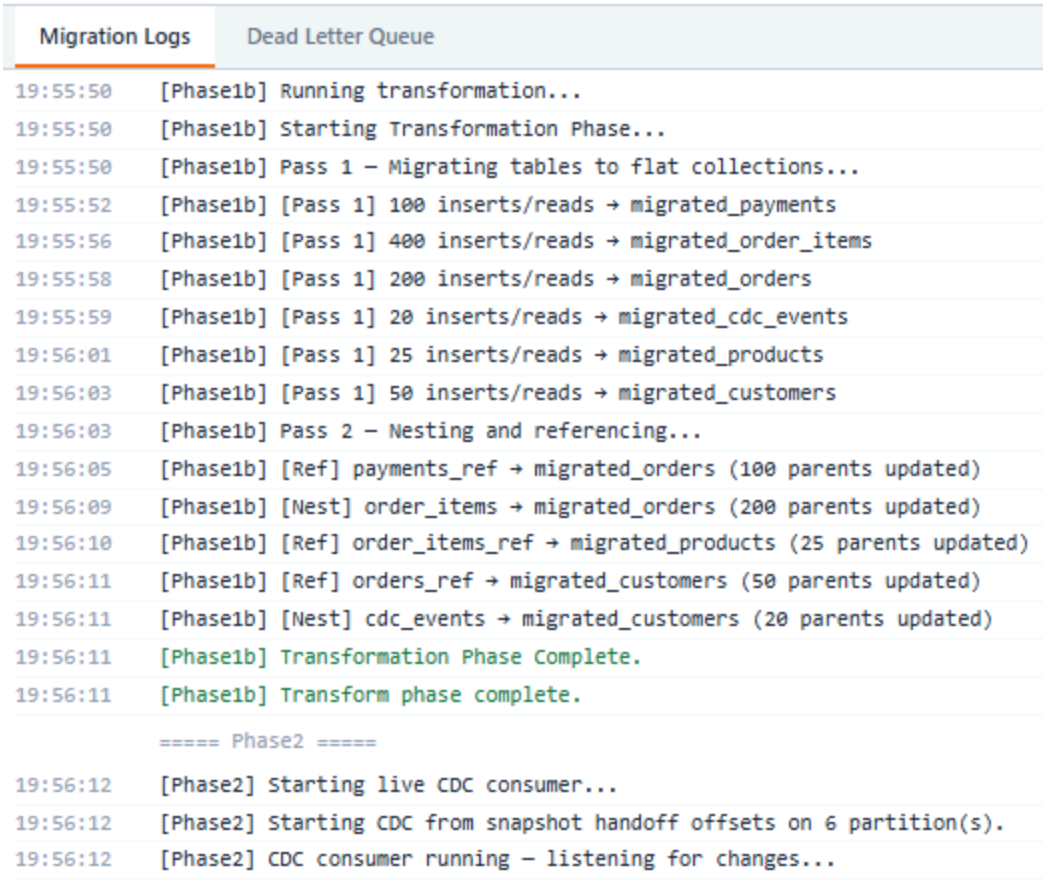


Figure 13. Migration logs showing phase, verification, and status messages.

9. MIGRATION CONTROLS

Control	When available	Effect
1. Pause	While running	Temporarily pauses migration processing.
2. Resume	While paused	Continues the paused migration.
3. Verify	During CDC	Pauses CDC and checks schema drift and data consistency. CDC resumes only if the result is clean.
4. Finish Migration	During CDC	Performs final verification. If clean, completes the provider-managed capture

		resources. migration and releases
5. Disconnect	While connected	Ends the active connection while keeping MongoDB data and resumable state.
6. Cancel	While running	Stops the migration and drops the entire configured MongoDB target database.

9.1. Pause and Resume

Select Pause to temporarily stop processing. The toolbar control changes to Resume. Select Resume to continue. History marks a user-paused migration as halted so it remains identifiable as recoverable.

9.2. Verify

Verify is available only during Phase2 (CDC). It temporarily pauses CDC, checks for breaking schema drift, and compares PostgreSQL with the staging data using chunk-level Merkle verification.

- Clean result: CDC resumes automatically.
- Issues found: CDC remains paused so the user can investigate and resolve the problem before selecting Resume.

Review the logs for affected tables and mismatch information. Verification does not automatically repair data.

9.3. Finish Migration

Finish Migration is the successful completion path and is available only during CDC. It runs a final schema-drift and data-consistency check.

- If the result is clean, GRASYA stops live capture, releases the provider-managed resources, and finalizes embedded relationships. Staging data is retained by the current implementation.
- If the result is not clean, finishing is aborted. The migration remains available so the user can review the logs and resolve the issue.

IMPORTANT: Use Finish Migration when the target has passed verification and is ready for cutover. Do not use Cancel to represent successful completion.

9.4. Disconnect

Disconnect ends the active connection and stops the current migration worker. It removes temporary provider-managed connector resources but keeps the PostgreSQL replication slot and MongoDB data needed for a later resume.

Use Disconnect when the run must be stopped without discarding its target data. The migration appears as halted in History when disconnected mid-run.

9.5. Cancel

Cancel is destructive. The confirmation dialog warns that GRASYA will stop the run, release its migration resources, and permanently drop the entire configured MongoDB target database.

Select Yes, Cancel Migration only when the run and all data in that target database are no longer needed. A cancelled migration cannot resume from its previous state; choosing Resume on its history entry prepares a fresh run.

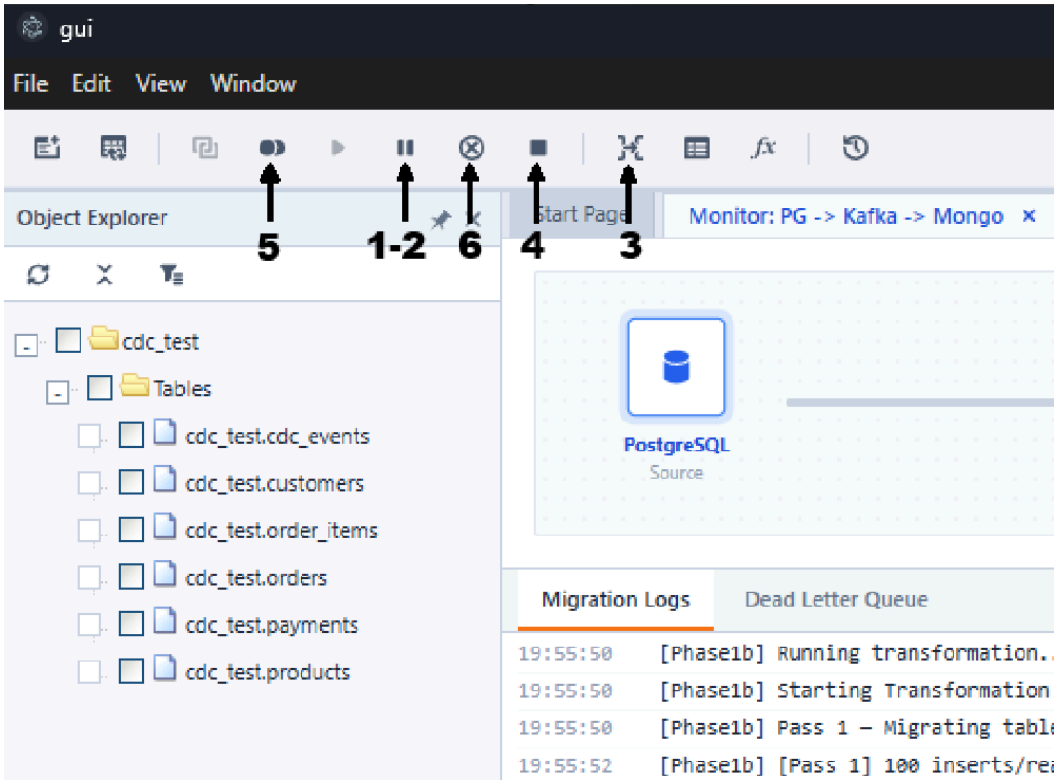


Figure 14a. Migration controls

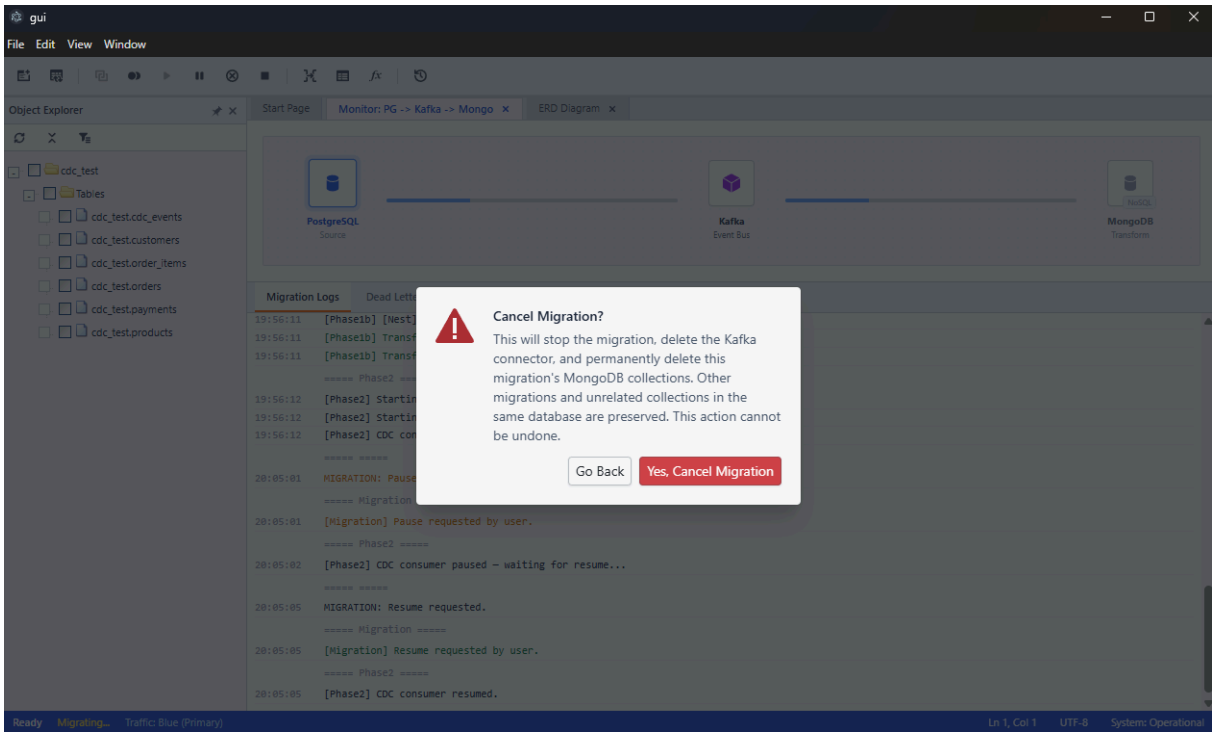


Figure 14b. Destructive Cancel Migration confirmation.

9.6. Why a control is disabled

- Connect is disabled after the source and target are connected.
- Run is disabled until connected and while a migration is already running.
- Pause/Resume and Cancel are disabled when no migration is running.
- Verify and Finish Migration are disabled until the migration reaches CDC.
- Relationship Apply and Calculated Fields Apply are disabled while running.
- Verification and finalization controls remain disabled while another verification operation is in progress.

10. ERD, LOGS, DLQ, AND MIGRATION HISTORY

10.1. ERD Diagram

Open the ERD Diagram workspace tab to inspect detected tables, columns, and foreign-key lines. Use it together with the MongoDB Relationship Mapping panel to decide how related data will be stored.

The ERD reflects the connected schema; it is not a general-purpose database editor. GRASYA does not require the user to alter source tables through this view.

10.2. Logs

The live Logs tab in Monitor is the first place to check when an operation is slow, unavailable, or unsuccessful. Logs include connection events, migration phase changes, relationship and calculated-field results, verification results, and lifecycle actions.

History can also retain logs associated with a saved desktop migration entry. These stored logs remain useful after a migration has stopped.

10.3. Dead Letter Queue (DLQ)

Open the DLQ tab from the Monitor view. A DLQ record represents a source event that GRASYA could not process successfully.

The list includes available values such as time, phase, topic, partition, offset, and failure reason. Select a row to expand its exception type and captured key/value details.

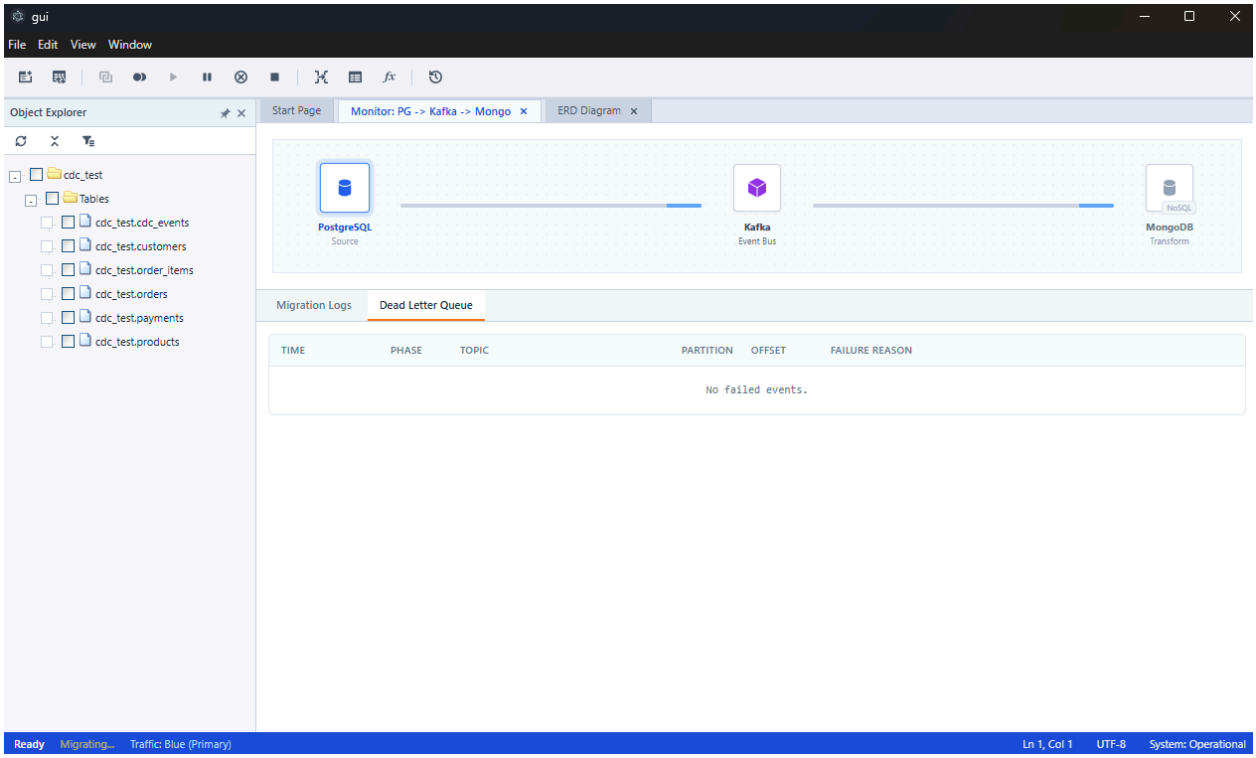


Figure 15. Dead Letter Queue tab without failed events.

IMPORTANT: The current application displays DLQ records but does not provide an automatic Replay button. Record the error, correct its underlying cause, and follow the recovery procedure supplied by the GRASYA administrator.

Captured DLQ content may contain source data. Handle screenshots and exported details according to the organization's data-handling rules.

10.4. Migration History

Select Migration History from the toolbar. History is stored locally by the desktop application and is available in the packaged Electron application. It is not available in browser-only development mode.

History entries may show these statuses:

- Running: migration is active.
- Finished: final verification succeeded and the migration was completed.
- Cancelled: the migration and target database were discarded.
- Halted: the run was paused, disconnected, interrupted, or otherwise left in a potentially resumable state.

- Failed: the run encountered an error.

Each entry displays source information, target database name, table count, and last update time. Available actions include:

- Logs: display logs stored for that migration.
- Resume: open the wizard with reusable settings prefilled.
- Delete: delete the local history entry.

Deleting a history entry removes the local record; do not treat it as a database-cleanup operation.

For security, History does not store the PostgreSQL password, replication password, or MongoDB URI. The user must enter these values again when resuming.

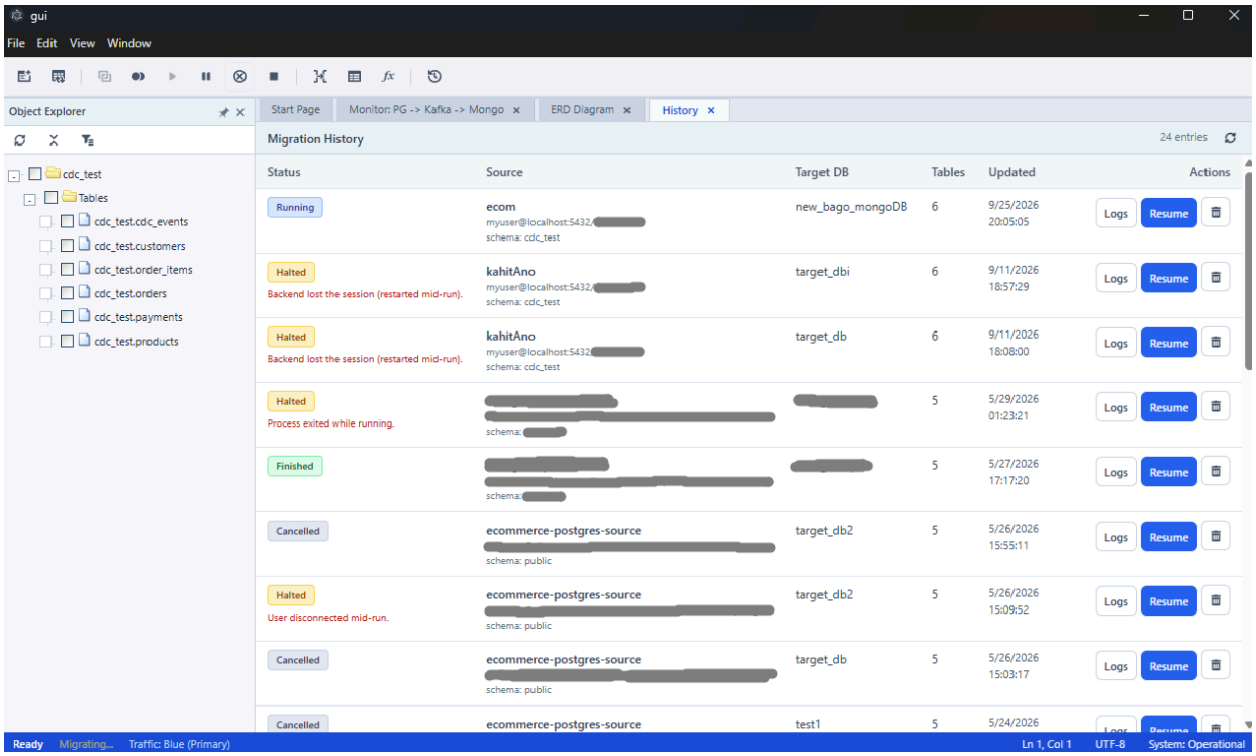


Figure 16. Migration History entries with Logs, Resume, and Delete actions. Some sensitive details are censored.

11. RESUMING AN INTERRUPTED MIGRATION

11.1. Resume procedure

To resume a halted migration:

1. Ensure the GRASYA cloud service is reachable.
2. Ensure no other migration is currently connected in the application.
3. Open Migration History.
4. Find the entry marked Halted.
5. Select Resume.
6. Review the prefilled source and capture settings.
7. Re-enter the PostgreSQL password and replication password.
8. Enter the MongoDB URI again. Credentials and URIs are not stored in History.
9. Proceed through the wizard and verify the target database and table list.
10. Select Finish to reconnect the configuration.
11. Review relationship mappings and calculated fields.
12. Select Run.

A true halted-run resume skips the original snapshot, re-runs the transformation against preserved staging data, and then continues CDC from the preserved source position.

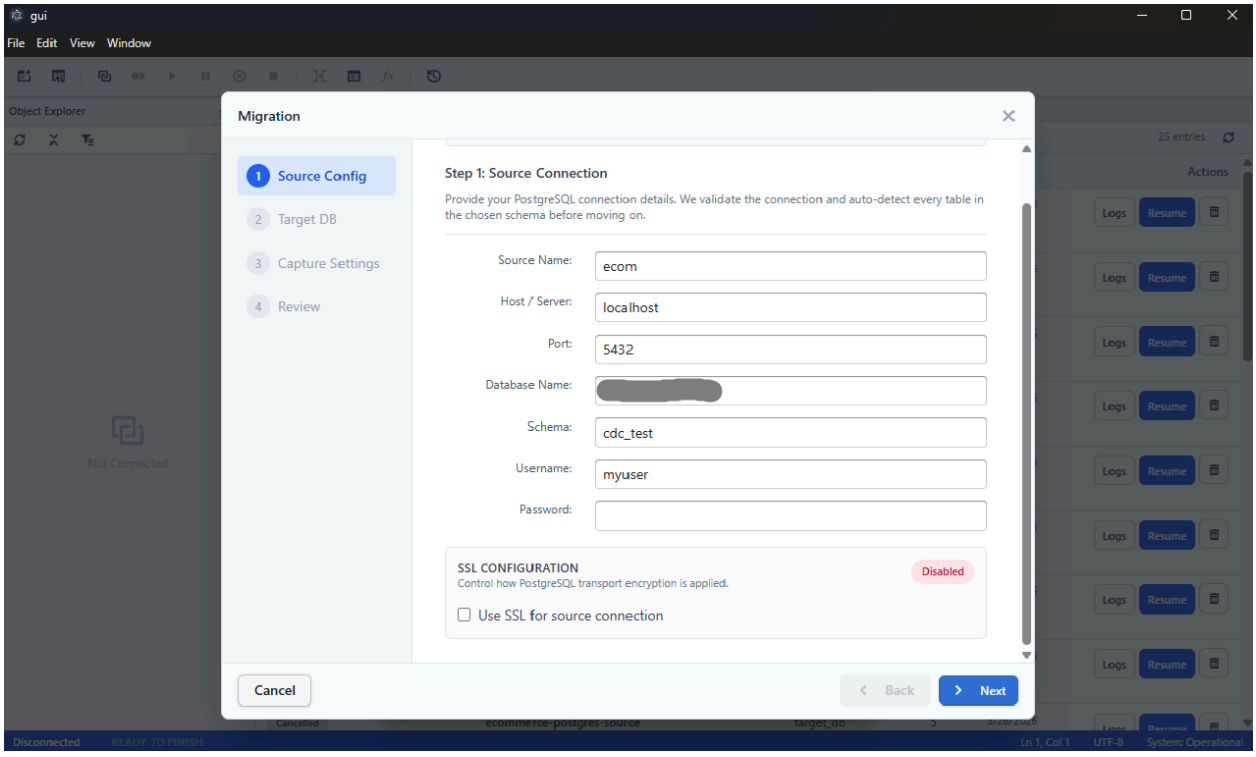


Figure 17. Resuming a halted migration. Saved settings are prefilled, while passwords and the MongoDB URI must be re-entered. A sensitive detail is censored.

11.2. Cancelled, finished, failed, and running history entries

Only a Halted history entry is treated as a resumable migration by the current application. Selecting Resume on a Cancelled or Finished entry prepares a fresh migration because the earlier replication state has already been released. Failed and stale Running entries are also treated as fresh runs by the current interface.

Review the wizard carefully before starting any history-based fresh run. An initial snapshot may copy all existing source rows again.

11.3. Missing replication slot

If a halted migration's preserved PostgreSQL replication slot no longer exists, GRASYA cannot continue from its previous source position. The wizard displays an explanation and offers Restart as fresh migration.

Use Restart as fresh migration only after confirming that a new full snapshot is appropriate. GRASYA changes the run to initial snapshot mode and starts from the beginning rather than claiming to resume the missing state.

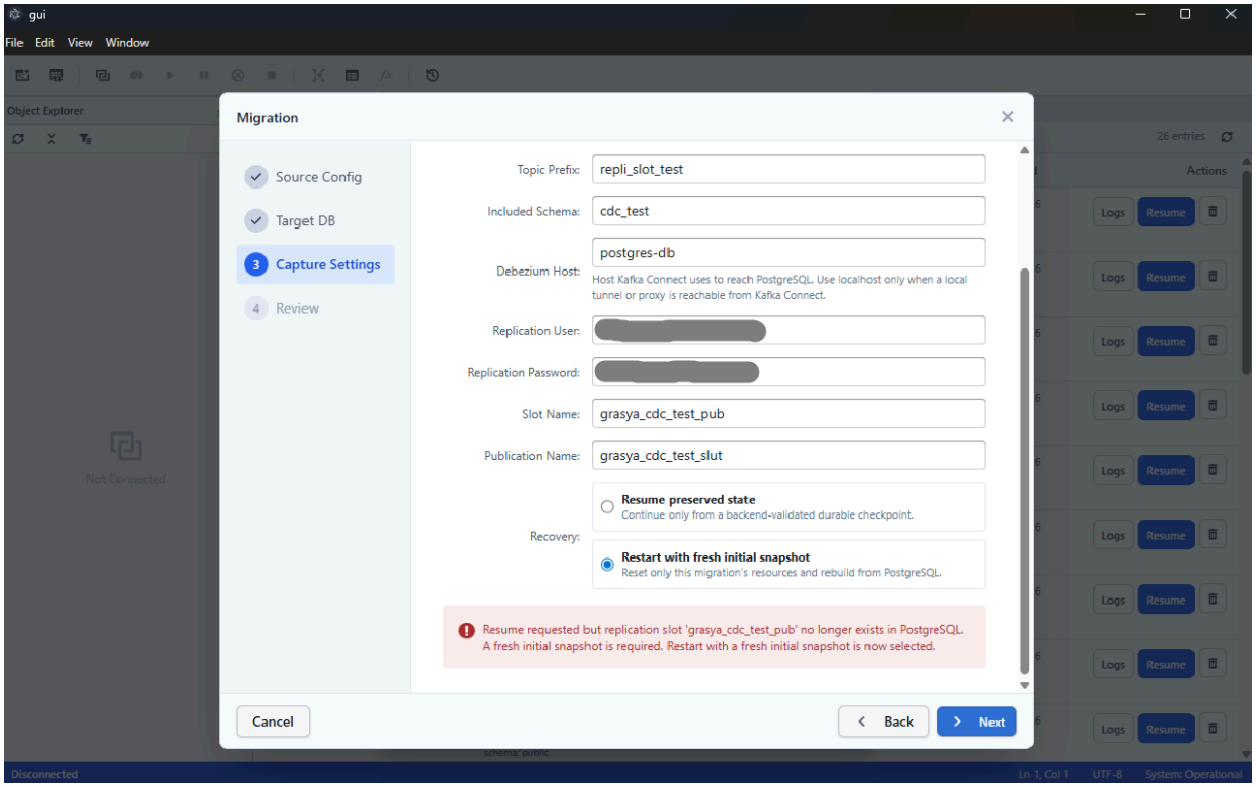


Figure 18. Restarting as a fresh migration when the saved replication state is no longer available. Some sensitive details are censored.

12. TROUBLESHOOTING AND COMMON ERRORS

12.1. GRASYA cannot reach the cloud service

Symptoms:

- Red status bar.
- No connection — attempting to reconnect message.
- No connection to server. Try again later message.

Actions:

1. Confirm the computer has internet or approved VPN access.
2. Check whether a firewall or proxy is blocking the GRASYA application.
3. Wait for the retry countdown.
4. If retries are exhausted, select Retry.
5. If a migration was active, check History after service access returns. The application may mark the interrupted run as Halted.
6. Contact GRASYA support if the service remains unavailable.

12.2. PostgreSQL connection fails

Check:

- Hostname, port, database, schema, username, and password.
- Whether SSL must be enabled or disabled.
- Whether the PostgreSQL server accepts connections from the GRASYA cloud service rather than only from the local desktop.
- Firewall, trusted-IP, VPN, or private-network rules.
- Whether the database account has login and schema-read permissions.

Do not repeatedly guess credentials. Ask the PostgreSQL administrator to confirm the account and review server-side connection logs.

12.3. MongoDB connection fails

Check:

- The full MongoDB URI and target database name.
- Whether the URI password contains characters that require URI encoding.
- Authentication database and TLS options.
- MongoDB trusted-IP or firewall settings.
- Whether the MongoDB account has access to the target.

Do not post the complete MongoDB URI in chat messages, tickets, or screenshots. Provide a masked URI and the exact non-secret error text when requesting help.

12.4. No tables are detected

Check that:

- The correct database and schema were entered.
- The schema contains ordinary tables visible to the supplied user.
- The PostgreSQL account can read schema metadata.
- Letter case and spelling match the real schema name.

The wizard cannot finish when no tables are detected.

12.5. Capture or permission error

Ask the PostgreSQL administrator to confirm:

- Logical replication is available and enabled.
- The replication username and password are correct.
- The account can create or use the configured slot and publication.
- The slot name is not in use by another active migration.
- The publication and included schema values are correct.

The client should not attempt to repair provider-managed Kafka or Debezium services. Escalate service-side errors to GRASYA support.

12.6. Missing replication slot during Resume

The earlier change-stream position cannot be recovered from the history entry alone. Choose Restart as fresh migration only if a new initial snapshot is acceptable. Otherwise, stop and consult the database administrator before continuing.

12.7. DLQ entries appear

1. Open Monitor and select DLQ.
2. Select an entry to view its reason and exception type.
3. Check the normal logs for related schema or transformation errors.
4. Record the time, phase, affected table/topic name, and error text.
5. Correct the underlying schema, permission, data, or expression problem with the appropriate administrator.
6. Follow the recovery procedure provided by GRASYA support.

Do not assume that displaying a DLQ record means it was retried. The current desktop application has no automatic DLQ replay control.

12.8. Verification reports schema drift or mismatched data

Verify may report breaking schema drift or a Merkle mismatch for one or more tables. When this occurs:

1. Leave the migration paused.
2. Read the verification and per-table log messages.
3. Check whether the PostgreSQL schema changed during migration.
4. Check the DLQ for source events that failed to apply.
5. Where possible, coordinate a stable source-write window and verify again after the identified problem is resolved.
6. Contact the database administrator or GRASYA support before finalizing.

Merkle verification identifies mismatched chunks; it does not automatically identify or repair the exact row. Do not use Finish Migration until the final verification is clean.

12.9. A toolbar button is disabled

- Connect requires a working cloud session and no active connection.
- Run requires a completed connection wizard.
- Pause/Resume and Cancel require a running migration.
- Verify and Finish require Phase2 (CDC).
- Relationship and calculated-field settings must be applied before Run.

- Controls may remain temporarily disabled during verification.

Use the Monitor logs and bottom status bar to confirm the current state.

12.10. Migration does not resume after an application or backend restart

Open History and use the Halted entry rather than creating unrelated settings from memory. Re-enter all omitted credentials. If GRASYA reports that the replication slot is missing, follow Section 11.3. If the target or source is no longer reachable, restore database connectivity before trying again.

13. DATA-SAFETY WARNINGS AND GLOSSARY

13.1. Data-safety checklist

Before connecting:

- Use a dedicated MongoDB target database.
- Verify all hostnames, database names, schemas, and ports.
- Confirm cloud-to-database network access with the administrators.
- Use least-privilege database accounts that still support the migration.
- Keep credentials out of documentation and screenshots.

Before selecting Run:

- Review all automatically detected tables.
- Apply relationship mappings.
- Review generated and calculated fields.
- Confirm initial or no_data snapshot mode is intentional.
- Confirm the target has enough capacity for staging and migrated data.

Before selecting Verify or Finish Migration:

- Coordinate with application owners if a stable source-write window is required for final cutover.
- Review the DLQ and migration logs.
- Resolve schema drift and data mismatches.
- Confirm downstream users are ready for the migrated MongoDB structure.

Before selecting Cancel:

- Read the confirmation dialog completely.
- Verify the displayed target database name.
- Confirm that every collection in that database may be permanently deleted.
- Use Disconnect instead when the migration may need to be resumed.

13.2. Credential and screenshot safety

Never include the following in screenshots or shared documents:

- PostgreSQL passwords.
- Replication passwords.
- Complete MongoDB URIs containing credentials.
- Private keys, tokens, or TLS secrets.
- Sensitive source rows shown in expanded DLQ entries.

Mask these values before capturing a screenshot. If a secret is accidentally shared, notify the responsible administrator and rotate it according to the organization's security procedure.

13.3. Glossary

PostgreSQL
The relational source database from which GRASYA reads tables and changes.

MongoDB
The document-oriented target database to which GRASYA writes staging and migrated collections.

Schema
A named PostgreSQL namespace containing tables and other database objects.

Collection
A group of documents in MongoDB, similar in purpose to a relational table but using document-shaped records.

ERD (Entity-Relationship Diagram)
A visual representation of tables, columns, and relationships discovered in the PostgreSQL schema.

Embedded relationship
A MongoDB representation in which related child data is stored inside a parent document.

Referenced relationship
A MongoDB representation in which related documents remain in separate collections and are linked by identifier values.

Snapshot
The initial copy of existing PostgreSQL rows performed at the start of a normal full migration.

CDC (Change Data Capture)
The continuous processing of inserts, updates, and deletes that occur after the initial snapshot.

DLQ (Dead Letter Queue)
A MongoDB collection and Monitor tab containing events that GRASYA could not process successfully.

Schema drift
A difference caused when the PostgreSQL structure changes during or after the migration configuration, such as a changed or removed column.

Merkle verification
A chunk-based hash comparison used by GRASYA to determine whether PostgreSQL rows and staged MongoDB payloads are consistent.

Replication slot
A PostgreSQL object that preserves a logical change-stream position so GRASYA can continue reading changes and resume an interrupted migration.

Publication
A PostgreSQL logical-replication definition identifying changes made available to the capture process.

Migration history
The local desktop record of past and active migration configurations, statuses, and stored logs. It intentionally excludes database passwords and MongoDB URIs.

Staging
The internal MongoDB area where GRASYA keeps a row-shaped source mirror for transformation and verification.

GRASYA cloud service
The provider-operated backend that coordinates connection validation, migration processing, monitoring, and verification for the desktop client.

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